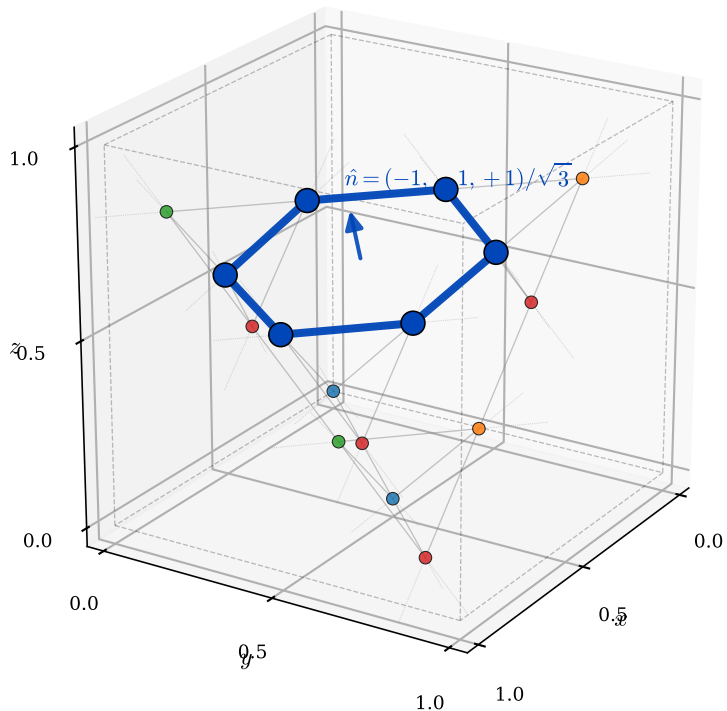


Two emergent ring-exchange loops on the $1\times 1\times 1$ pyrochlore cluster

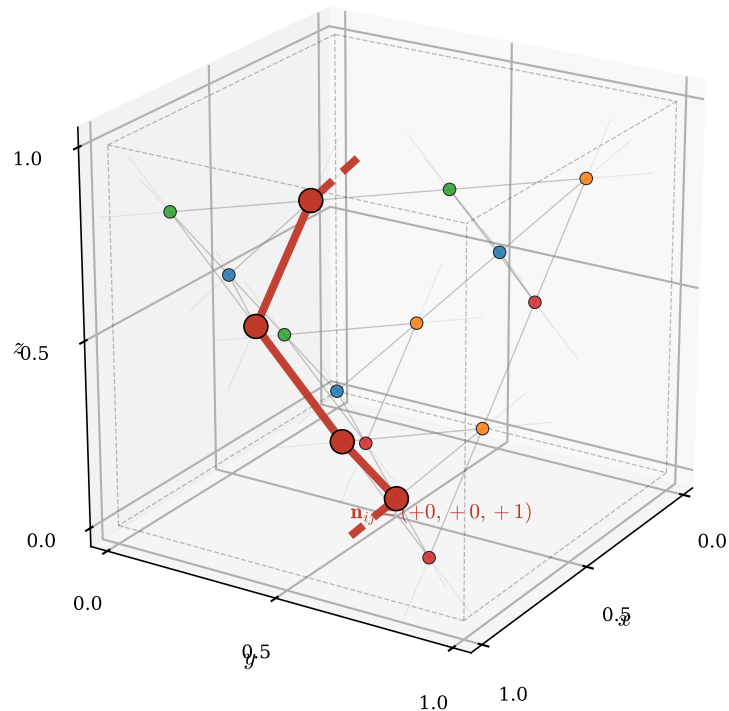
(a) Contractible hexagon $\mathbf{w} = 0$

$$g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2$$



(b) Wrapping 4-cycle $\mathbf{w} = (+0, +0, +1)$

$$g_4^{\text{spur}} \sim 4|J_{\pm}|^2/J_{zz}$$



Solid bonds: bulk ($\mathbf{n}_{ij} = 0$). Dashed grey bonds: boundary-crossing. Sublattices $\mu = 0/1/2/3 = \text{blue/orange/green/red}$.